

Neuroanatomy: Blood-Brain Barrier, Meninges, CSF, Ventricles

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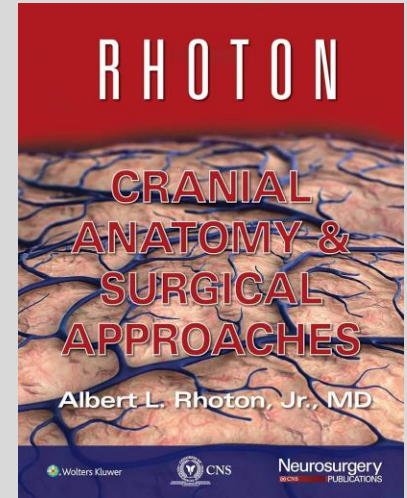
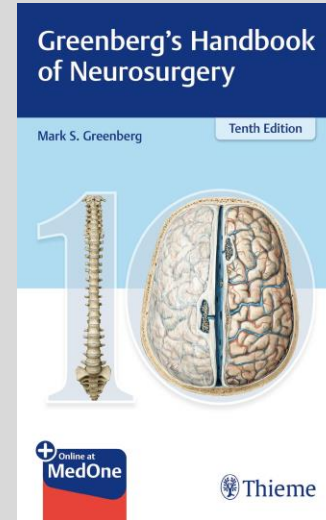
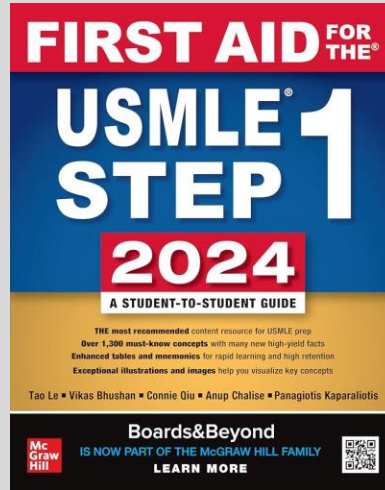
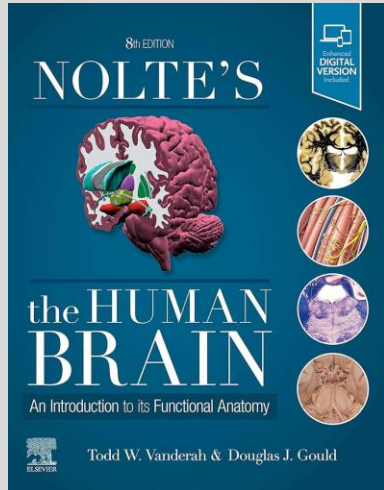
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References

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Session Objectives

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1. Explain the **three interphases that seals central nervous system** from the extracellular fluid and rest of the body
2. Identify and describe the major structural components and functional roles of the **blood-brain barrier**.
3. Explain and differentiate the anatomy of the **scalp, skull, meninges, meningeal spaces**, and mechanisms of common **brain herniations**.
4. Illustrate and label the morphology and key features of the **ventricular system**, including the **choroid plexus** and its pathologies such as **hydrocephalus**.
5. Analyze and predict how the **blood-brain barrier, cerebrospinal fluid**, and **ventricular system** operate in **maintaining neural homeostasis**, and assess the **functional consequences of deficits** in these structures due to injury or disease.

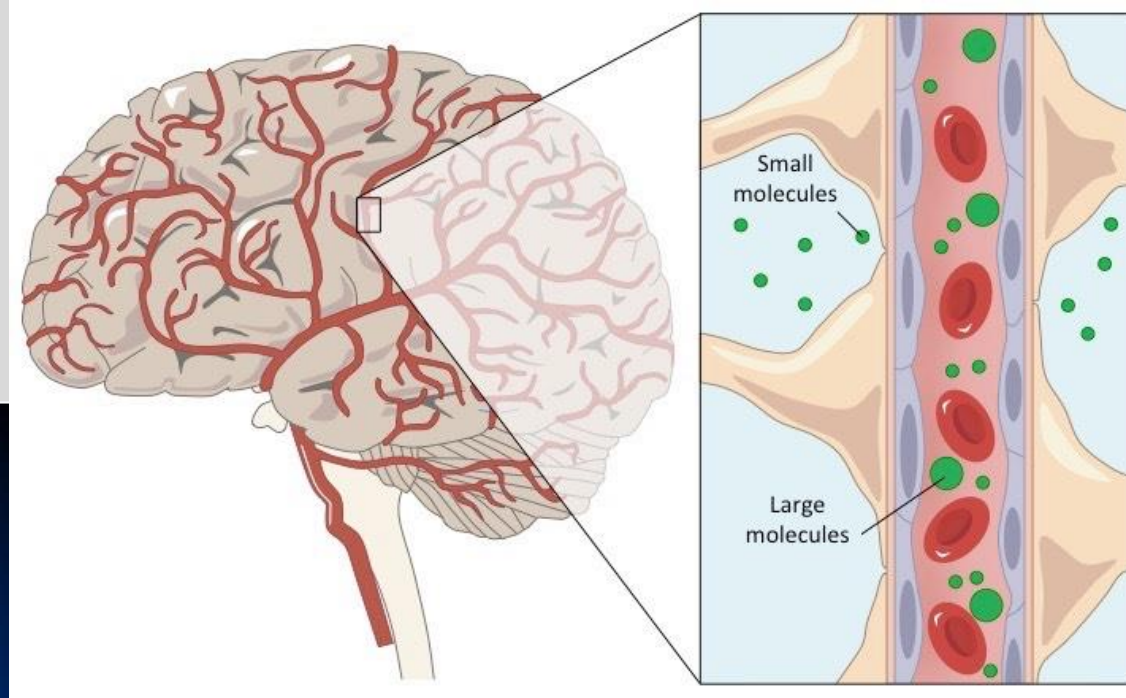
Clinical Question

A patient presents with neurological symptoms secondary to increased blood-brain barrier (BBB) permeability after a traumatic brain injury. Damage to which of the following proteins, essential in tight junctions, could disrupt BBB integrity by weakening cell-cell adhesion between endothelial cells?

- A.** Connexin-43
- B.** Laminin
- C.** Zona Occludens
- D.** Integrin

We'll come back to this soon!

Central Nervous System (CNS) Barriers

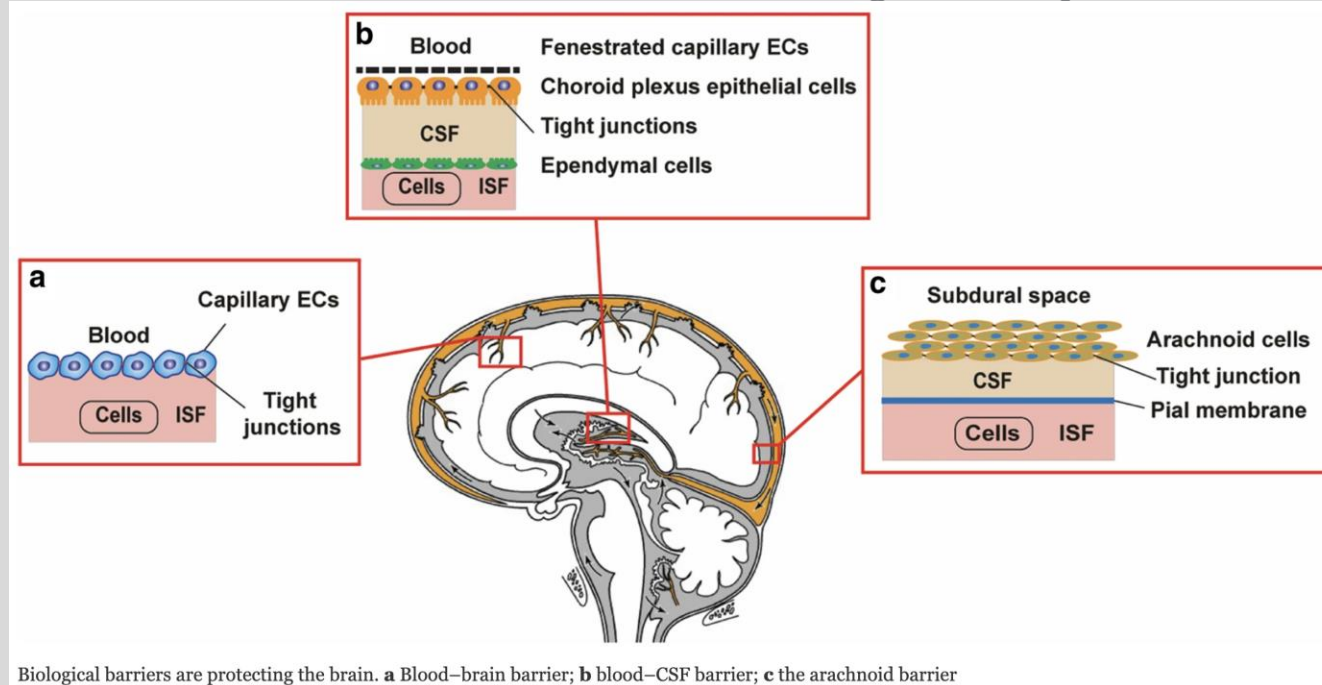


Central Nervous System (CNS) Barriers

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- a) **Blood-brain barrier:**
created by the endothelial cells that form the walls of the capillaries
- b) **Blood-CSF barrier:** formed by the **epithelial cells of the choroid plexus** facing the cerebrospinal fluid
- c) **Arachnoid barrier;**
provided by the **avascular arachnoid epithelium**, underlying the dura, and completely enclosing the CNS



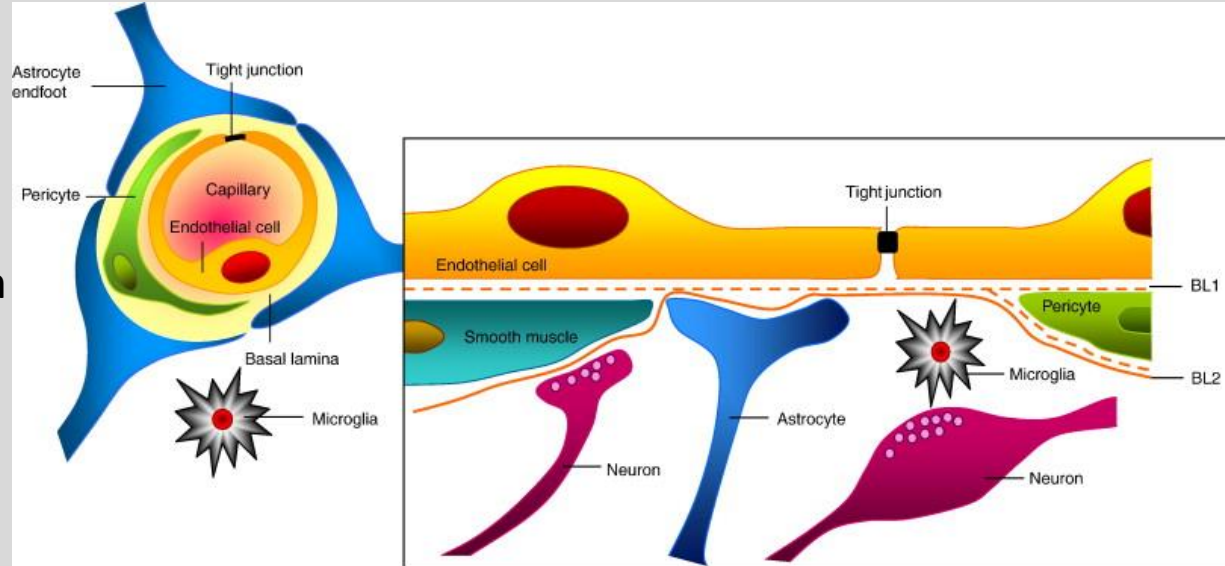
Biological barriers are protecting the brain. **a** Blood–brain barrier; **b** blood–CSF barrier; **c** the arachnoid barrier

Blood Brain Barrier (BBB)

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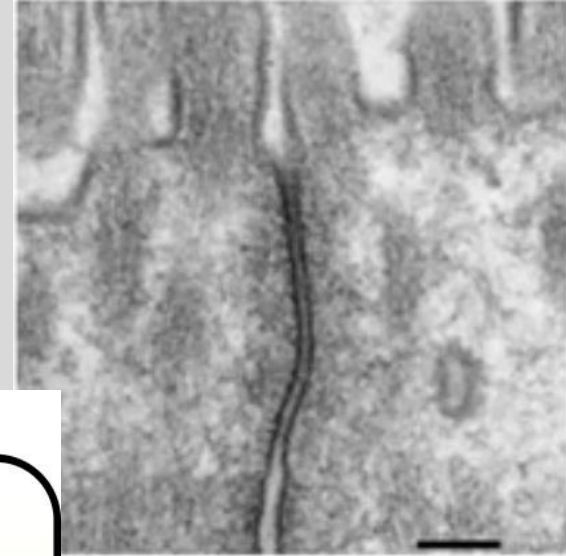
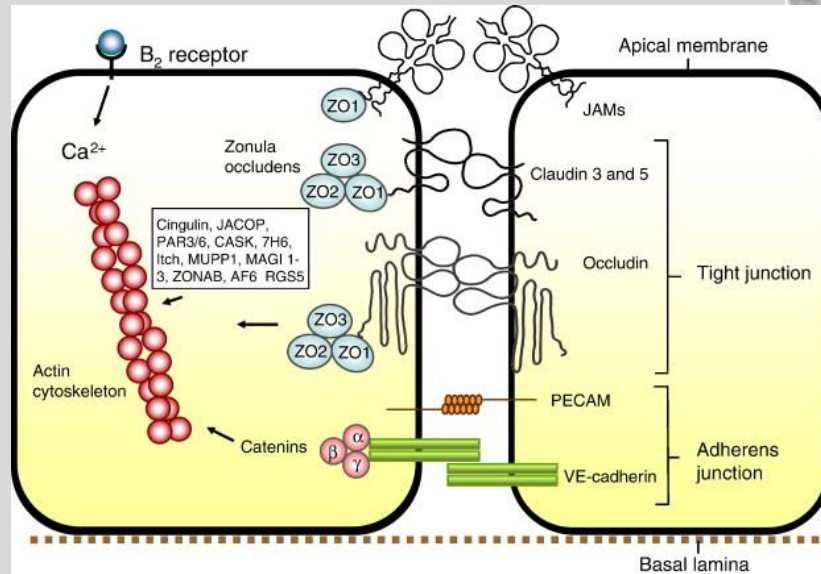
- A **selective semipermeable** membrane between the **blood** and the **interstitium of the brain**
- Allows the cerebral blood vessels to regulate movement of molecules and ions between blood and brain.
- Composed of **3 major structures**:
 - **endothelial cells (ECs)**
 - **pericytes (PCs)**
 - **astrocyte end-feet**



Blood Brain Barrier (BBB)

Tight Junction (Zona Occludens)

- Located **between cerebral endothelial cells**
- **Zona Occludens** connect the of **Occludin** and **Claudin** transmembrane proteins to the cytoskeleton.
- **Impenetrable to fluids, ions, and small molecules**
- **Essential for maintaining cell polarity.**

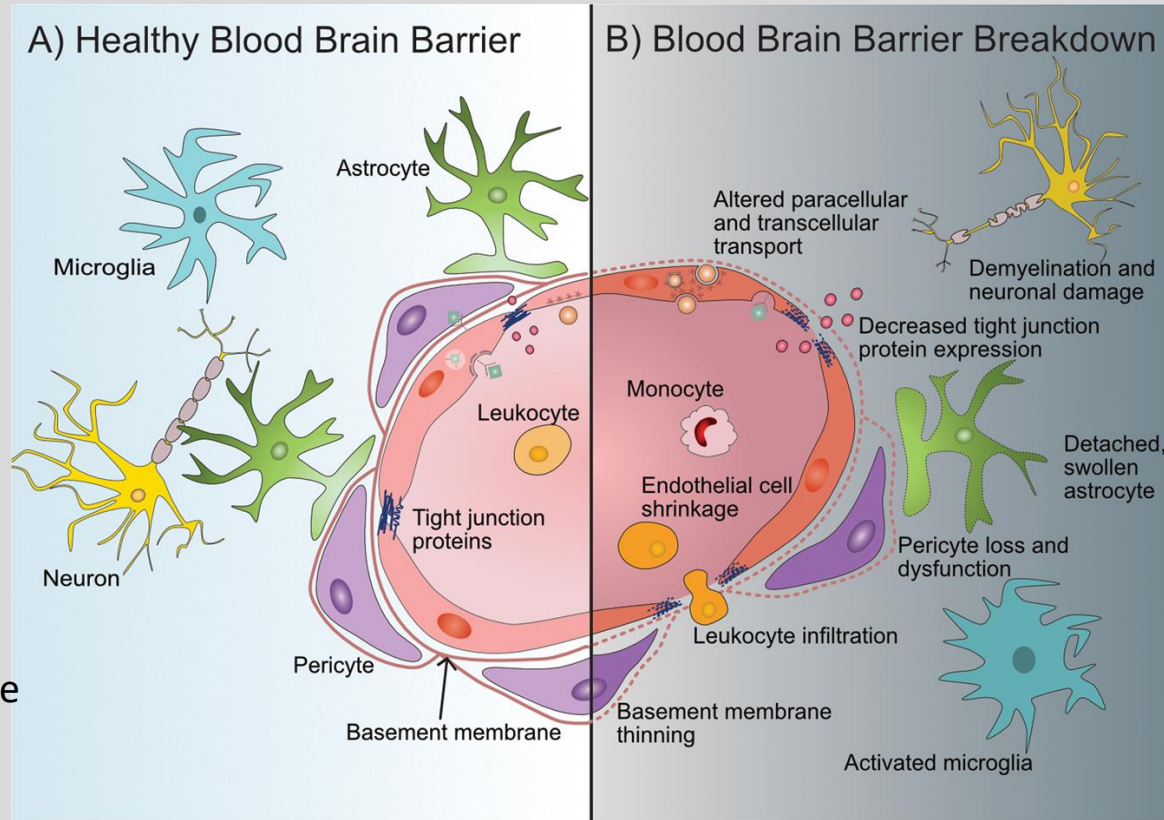


TEM of tight junction

BBB disruption during disease states

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- Disease states that alter BBB:
 - **Tumor**
 - **Traumatic brain injury**
 - Hypoxic-ischemic encephalopathy
 - HIV
 - **Multiple sclerosis**
 - **Alzheimer's disease**
 - Vasogenic edema
 - Hepatic encephalopathy
 - **Intraventricular hemorrhage**
- These pathological insults can alter the integrity of the BBB with consequent migration of leukocytes into the brain parenchyma



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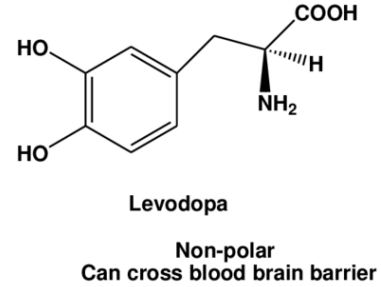
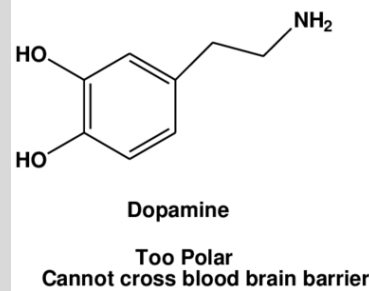
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Zona occludens proteins (ZO-1, ZO-2, ZO-3) are critical in forming and maintaining tight junctions in the BBB. Traumatic brain injury can disrupt these proteins, weakening cell-cell adhesion and increasing BBB permeability. This dysfunction allows immune cells and inflammatory molecules to enter the brain parenchyma, potentially contributing to neuroinflammation and worsening neurological outcomes.

Blood Brain Barrier (BBB)

- **Lipophilic barrier** → only lets in lipid substances (some exceptions)
- **Higher lipid solubility → higher BBB permeability**
- Very important for **pharmacology**
- Parkinson's disease → lack of dopamine in the brain; treatment targets to increase dopamine.
 - Dopamine is polar
 - Use Levodopa → non-polar; penetrate BBB
 - Once inside brain, Levodopa gets decarboxylated into Dopamine



Clinical Question

Levodopa, a primary treatment for Parkinson's disease, can frequently cause nausea as a side effect. This is due in part to levodopa's interaction with which structure in the brainstem, known for lacking a blood-brain barrier and for its role in detecting blood-borne emetic agents?

- A. Nucleus Tractus Solitarius
- B. Pineal Gland
- C. Area Postrema
- D. Subfornical Organ

We'll come back to this soon!

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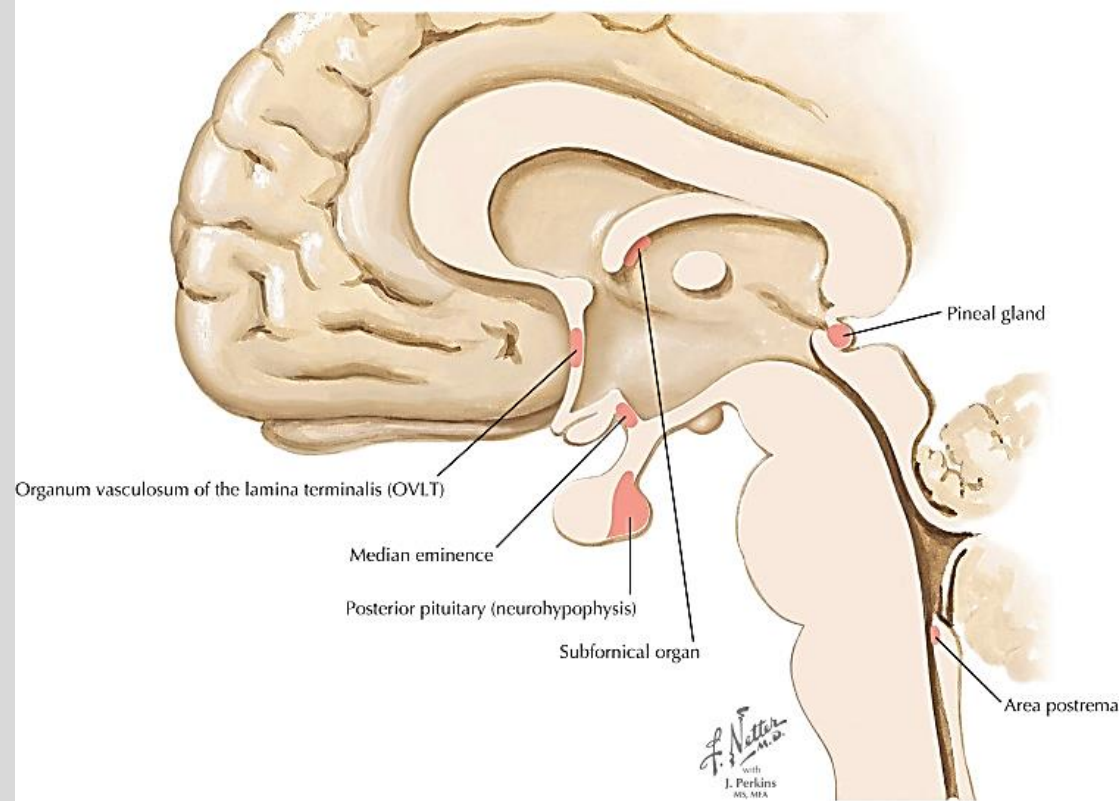
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Circumventricular Organs: Lacks BBB

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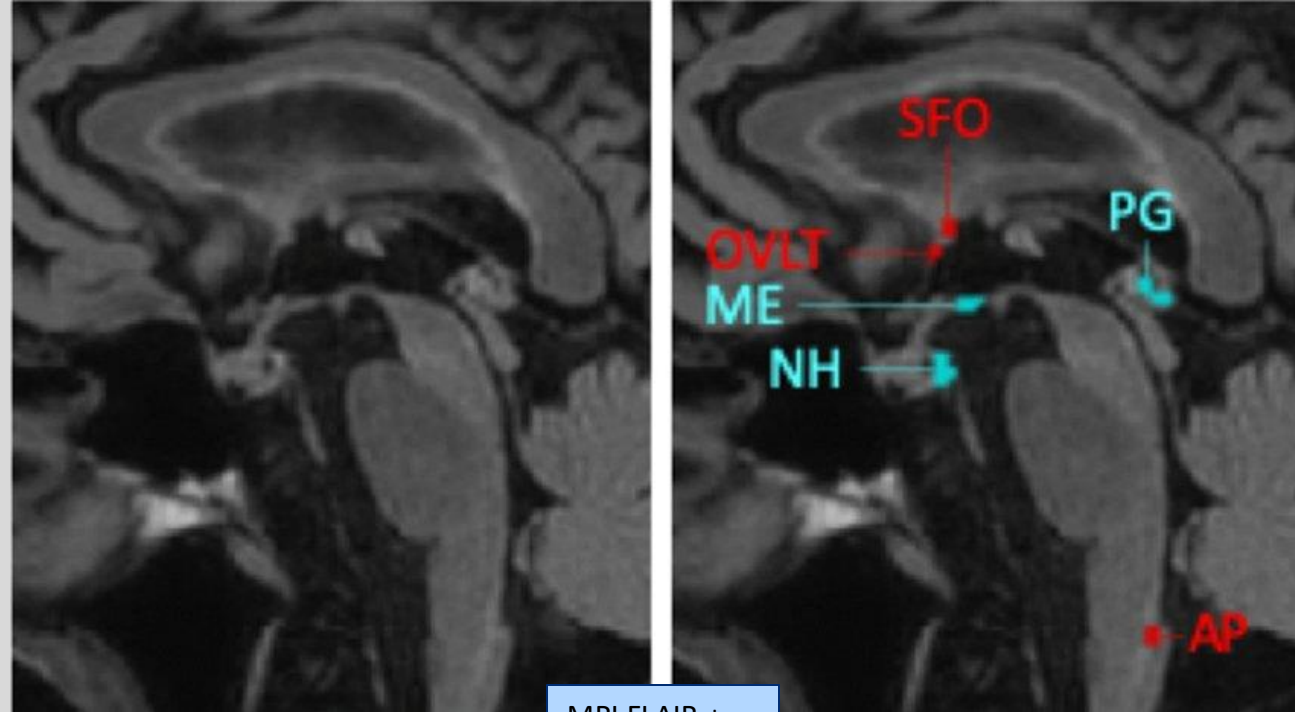
- The blood-brain barrier is present in all brain regions, except for **the circumventricular organs**:
 - Median eminence
 - Posterior pituitary (neurohypophysis)
 - Pineal gland
 - Subfornical organ
 - Organum vasculosum of the lamina terminalis
 - Area postrema- **Emetic center**
- Permits rapid transmission of biochemical influences



Circumventricular Organs (CVO): MRI w/ contrast

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- CVO are often visible in **contrast enhanced** imaging due to the lack of BBB
- Easy to mistake these areas as some kind of pathology if you do not know the anatomy!

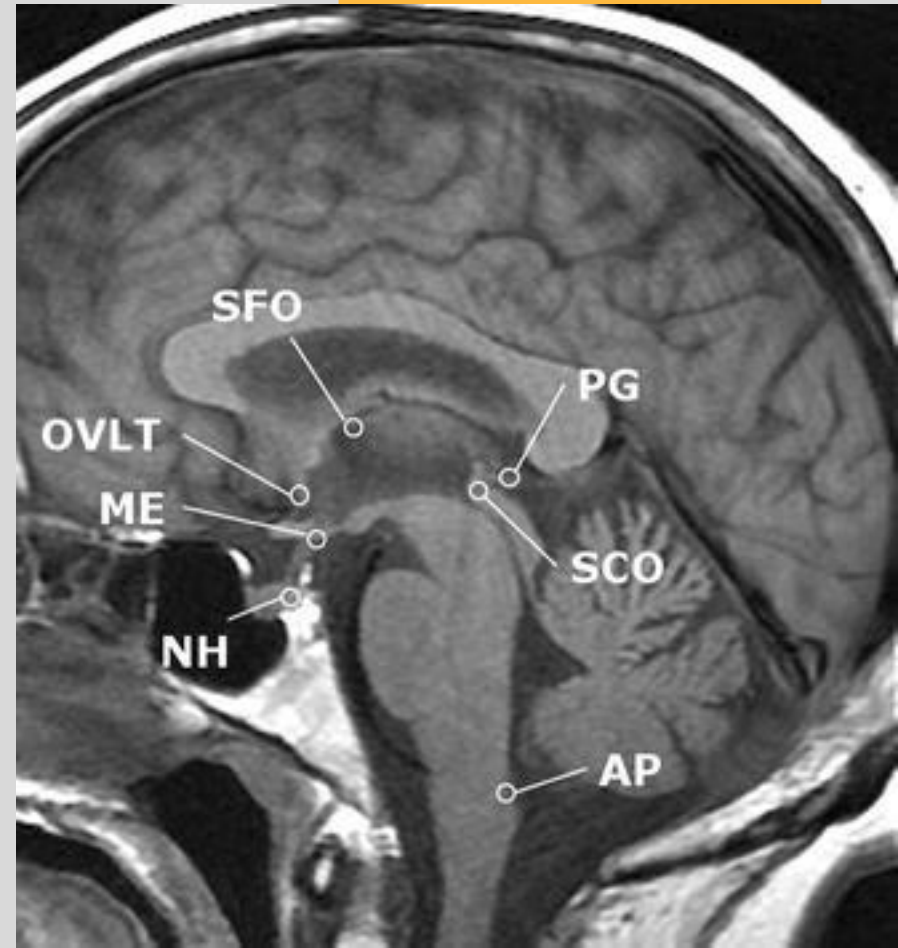


MRI FLAIR +

Verheggen ICM, de Jong JJA, van Boxtel MPJ, Postma AA, Verhey FRJ, Jansen JFA, Backes WH. Permeability of the windows of the brain: feasibility of dynamic contrast-enhanced MRI of the circumventricular organs. Fluids Barriers CNS. 2020 Oct 28;17(1):66. doi: 10.1186/s12987-020-00228-x. PMID: 33115484; PMCID: PMC7594295.

Area Postrema (AP): emetic center

- Located in the **caudal medulla**
- Molecules that are exposed in the plasma that are toxic, or emesis inducing can also gain quick access to area postrema → **cause vomiting reflex**
- Levodopa → frequently (approximately 30%) cause nausea and vomiting as unavoidable side effects by stimulating *dopamine receptors* in peripheral and brain tissues
 - Add carbidopa: decarboxylase inhibitor, prevents Levodopa conversion to dopamine, thus relieved the N/V induced by dopamine binding to AP



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Clinical Question

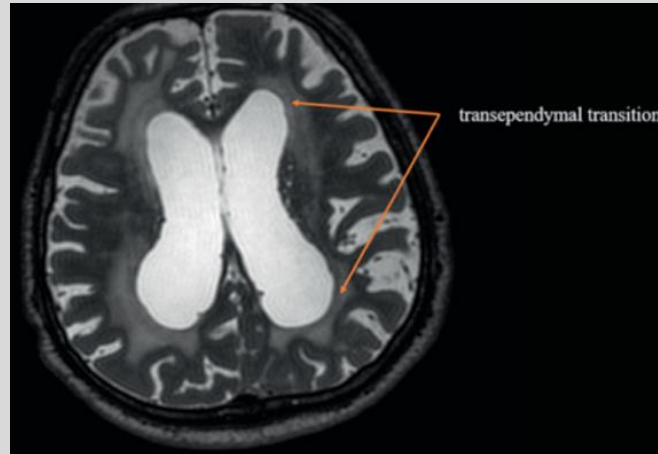
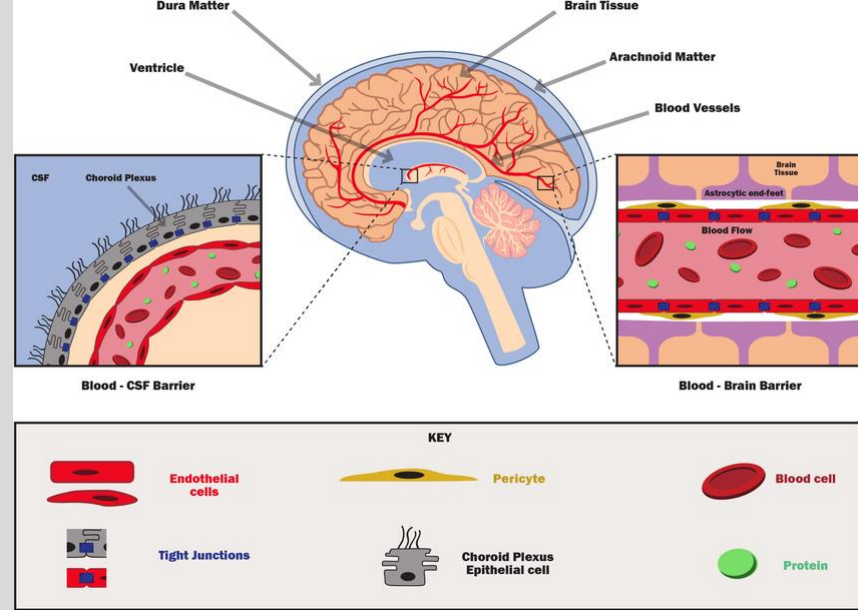
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The area postrema is sensitive to circulating blood-borne agents, such as levodopa, because it lacks a blood-brain barrier. This allows it to detect emetic substances directly in the bloodstream, often leading to nausea in patients taking dopaminergic medications for Parkinson's disease.

Blood-CSF Barrier

- **Blood-CSF barrier:** formed by the **epithelial cells of the choroid plexus (ependymal cells)** facing the cerebrospinal fluid
- In states of chronic **hydrocephalus:** the increased amount of CSF pushes on the endothelial cells and **stretches tight junction:** **leaky paracellular CSF flow** into the brain parenchyma: **Transependymal CSF flow**



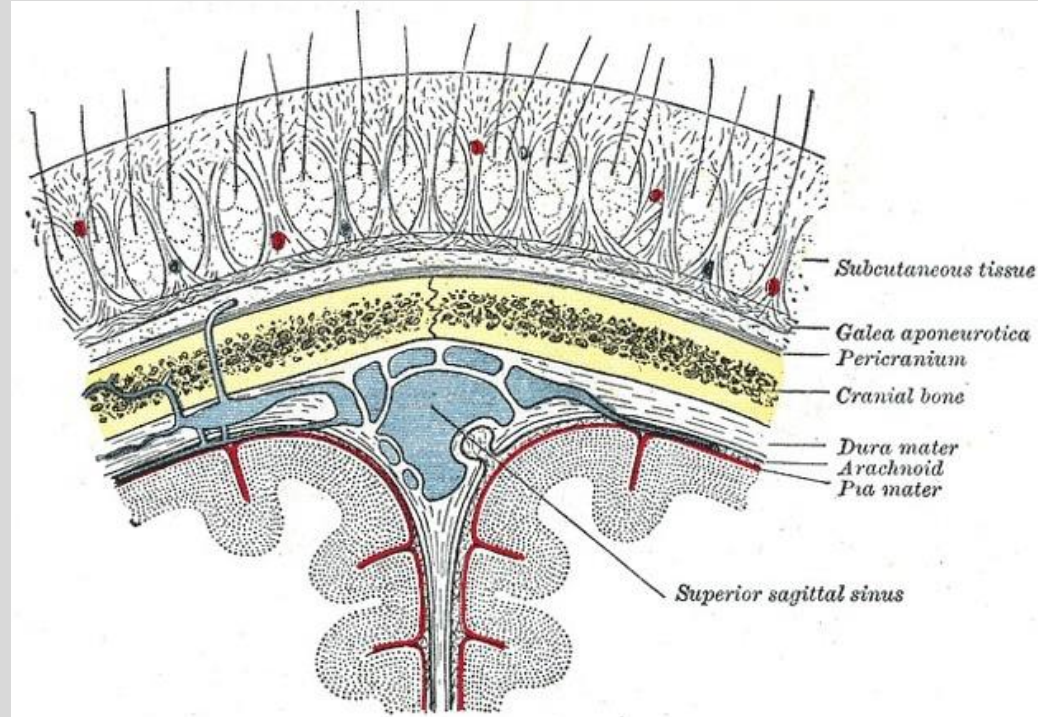
The Scalp

- The **scalp** is the skin and subcutaneous tissue covering the neurocranium.
- Composed of **five layers** and has a **rich vascular supply** which explains why it **bleeds profusely and heals well**.

1. **S: Skin** (epidermis, dermis)
2. **C: Connective tissue** (subcutaneous tissue)
3. **A: galea Aponeurosis**- strong layer, attachment for muscles of the scalp
4. **L: Loose connective tissue** (sponge-like areolar tissue) – potential
5. **P: Periosteum** (pericranium)- dense fibrous tissue

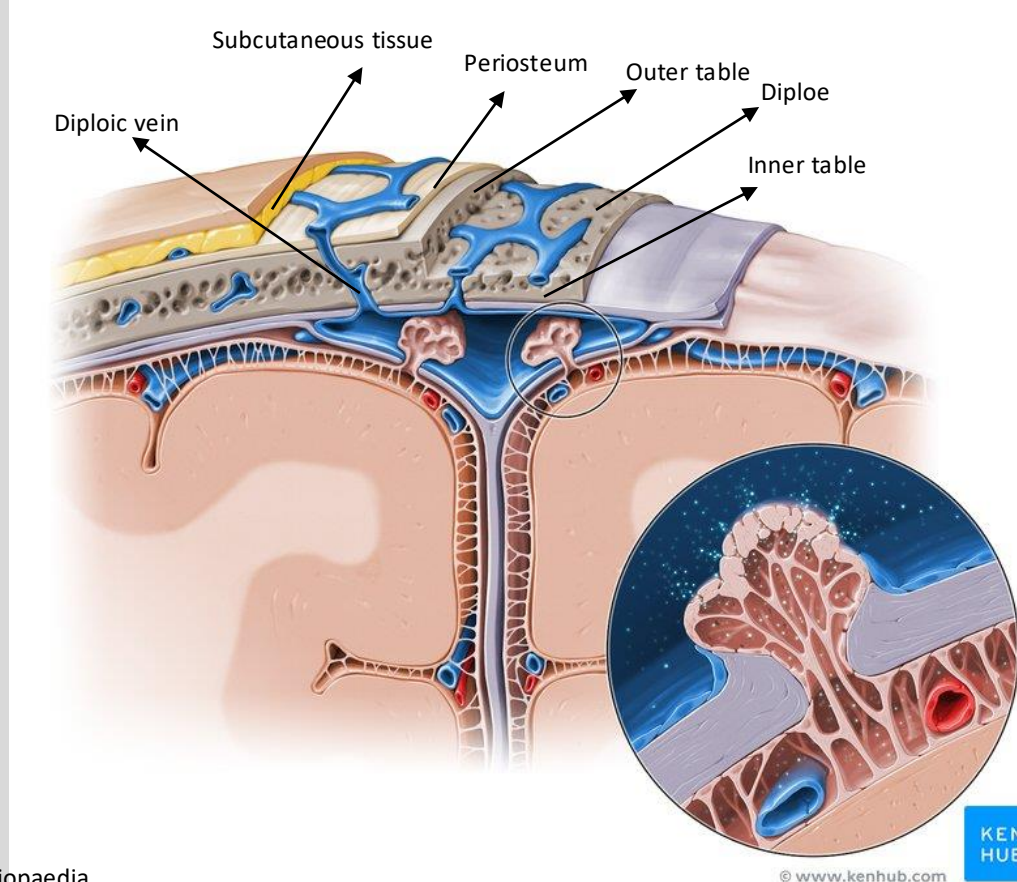
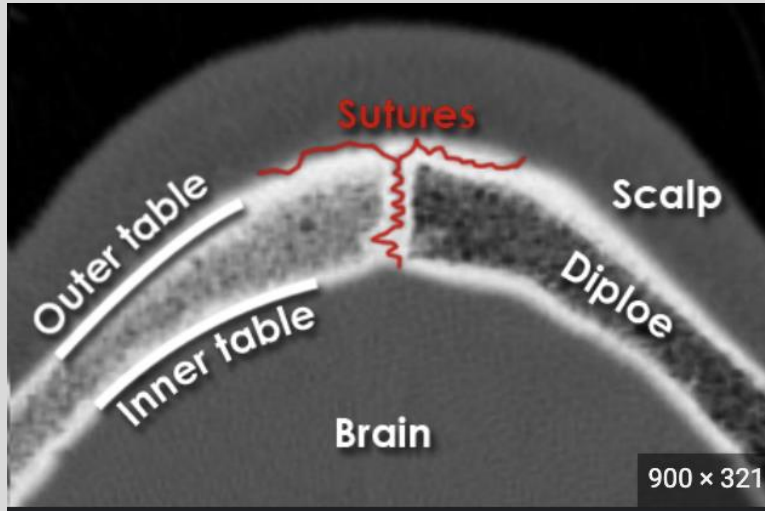
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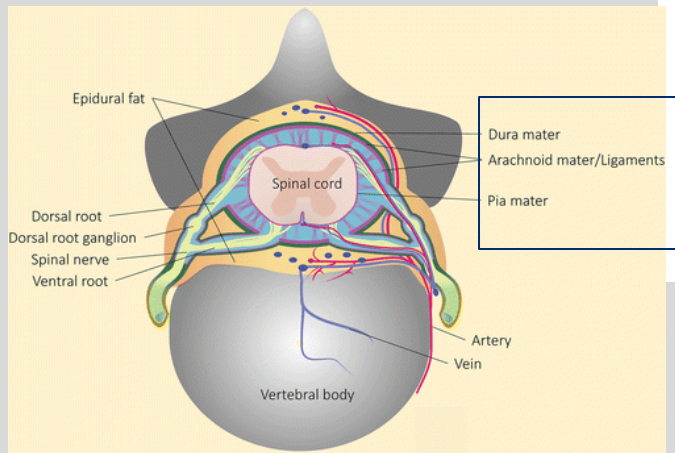
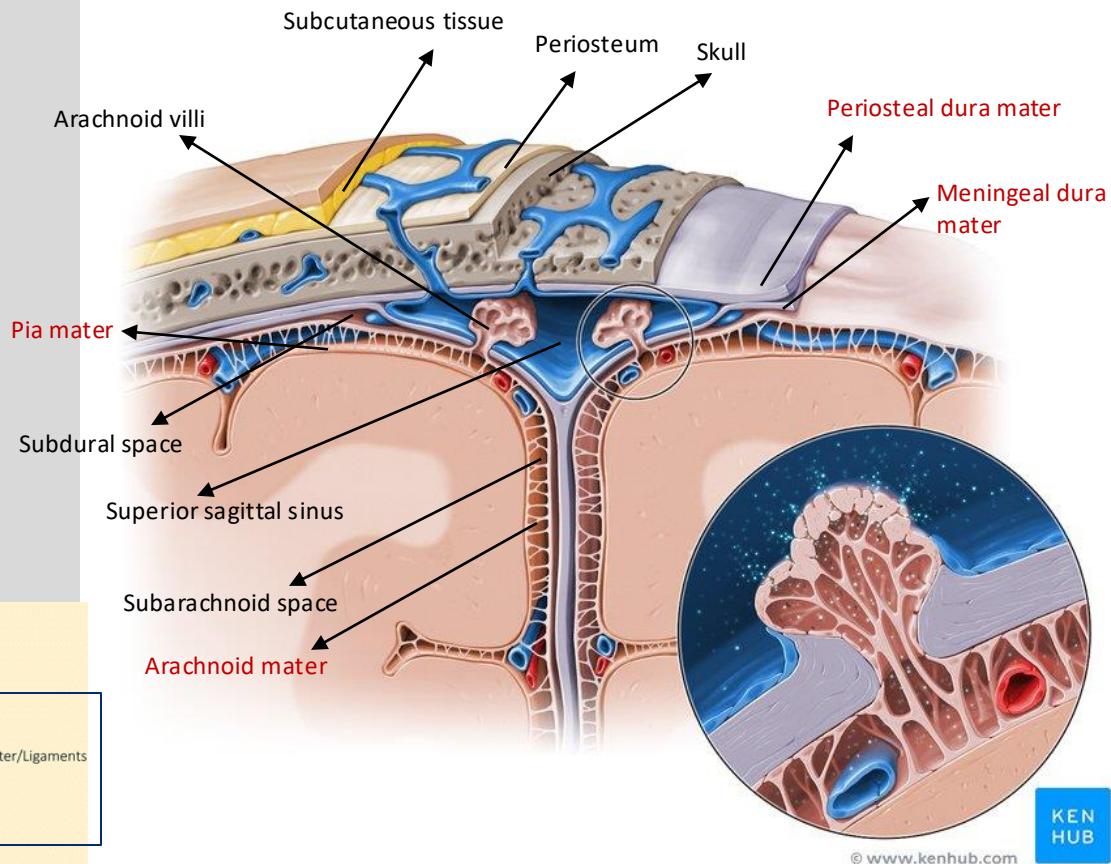
The Skull

The skull is composed of **flat bones**, with cortical bone forming the outer and inner tables and spongy bone creating the middle diploe layer: **“wafers-chocolate-wafers”**



The Meningeal Layer

- The brain and spinal cord are enveloped within **three layers of membrane collectively known as the meninges**
- **Superficial to deep, the three layers are the dura mater, arachnoid mater, and pia mater**
- The term “mater” is Latin for mother.

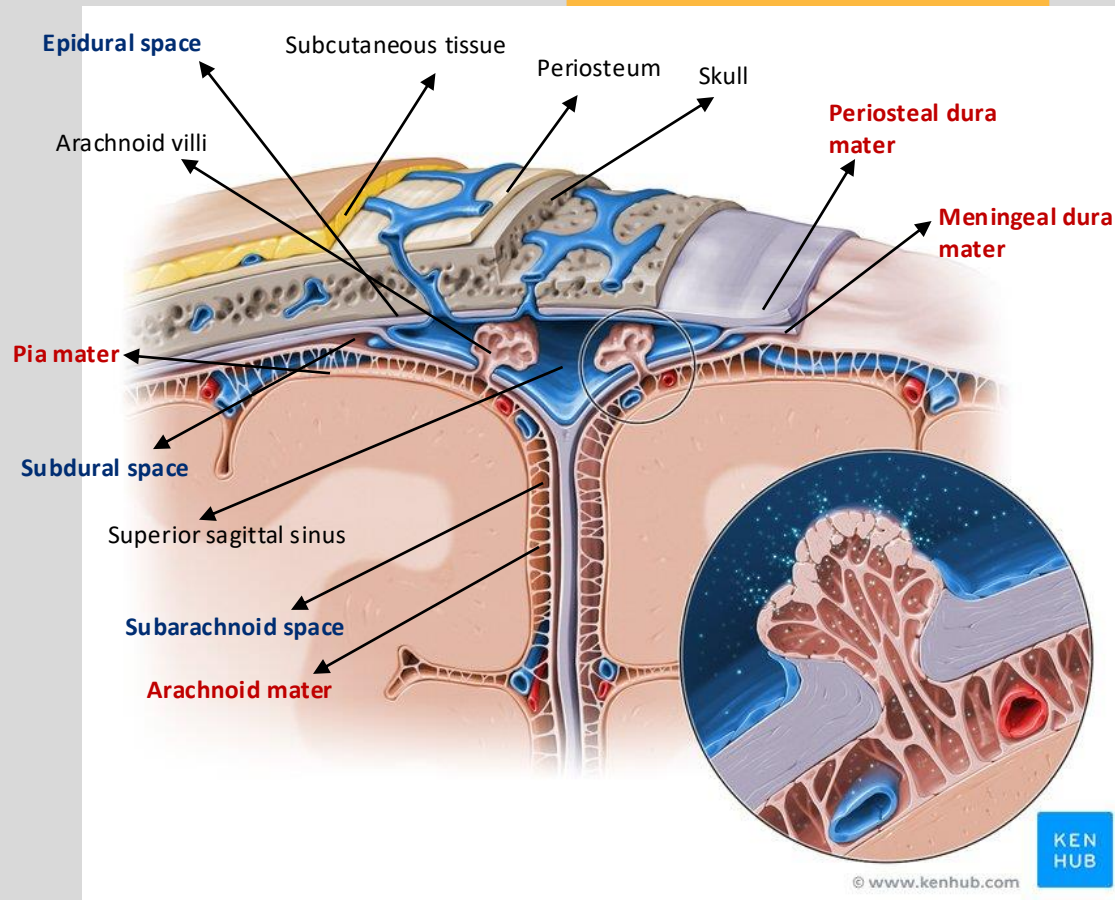


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The Meningeal Layer

1. The **dura** (hard) mater is composed of 2 layers of dense connective tissue and **adheres to the inner surface of the skull and vertebrae**.
 - **Periosteal dura**
 - **Meningeal dura/dura propria**
2. The **arachnoid** (spider like) mater is a thin, wispy membrane that lies just **deep to the dura and is superficial to the pia mater**.
3. **Pia** (soft) mater is a very thin, clear membrane that directly adheres to the surface of the brain and spinal cord.



The Meningeal Layer

Epidural Space

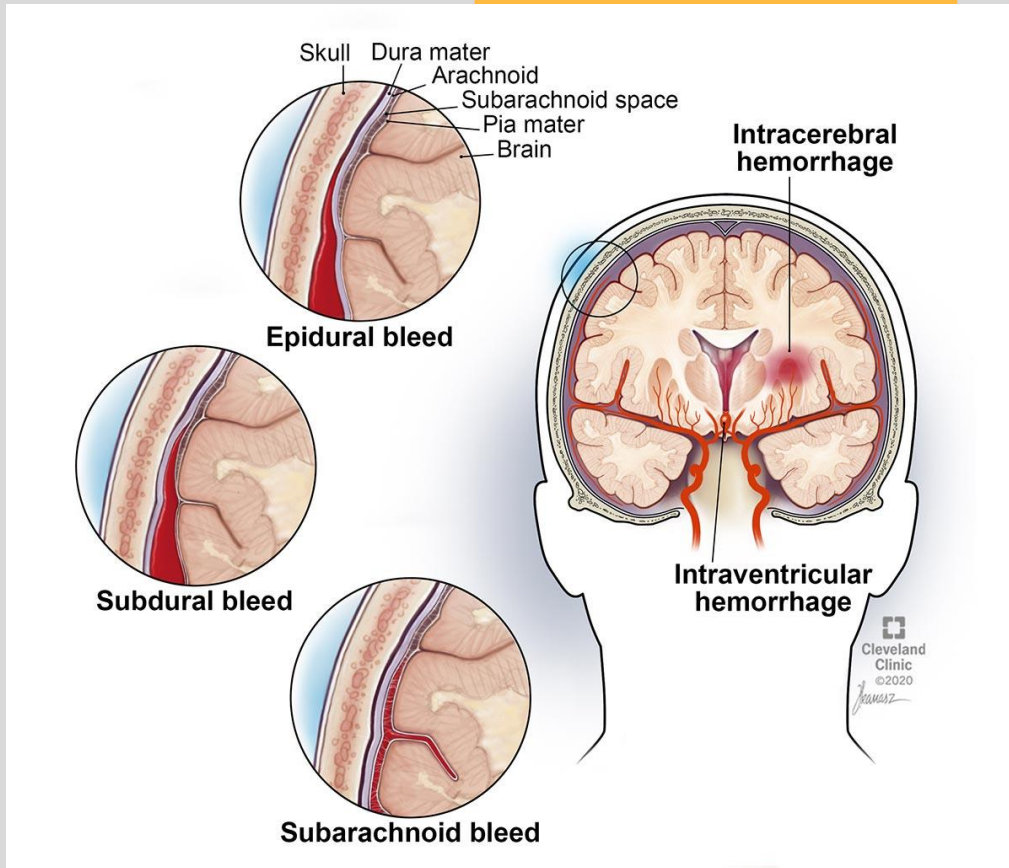
- Potential space between the inner skull and dura mater
- Contains **Middle meningeal artery**, which when injured (trauma) causes **epidural hematoma**

Subdural Space

- Between dura mater and arachnoid mater
- Contains **bridging veins**.
- **Subdural hematoma** occurs often from tearing of these **bridging veins**.

Subarachnoid Space

- Between **arachnoid mater** and **pia mater**
- Contains **Cerebrospinal fluid (CSF)** and major cerebral arteries
- **Subarachnoid hemorrhage**; Often from **aneurysm rupture**



More on this in the Cerebrovascular anatomy lecture..

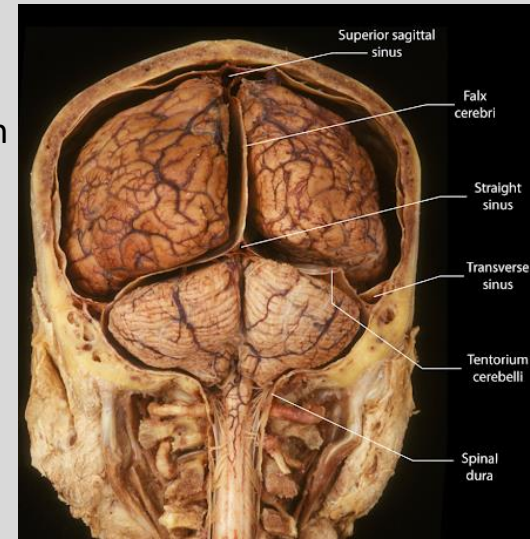
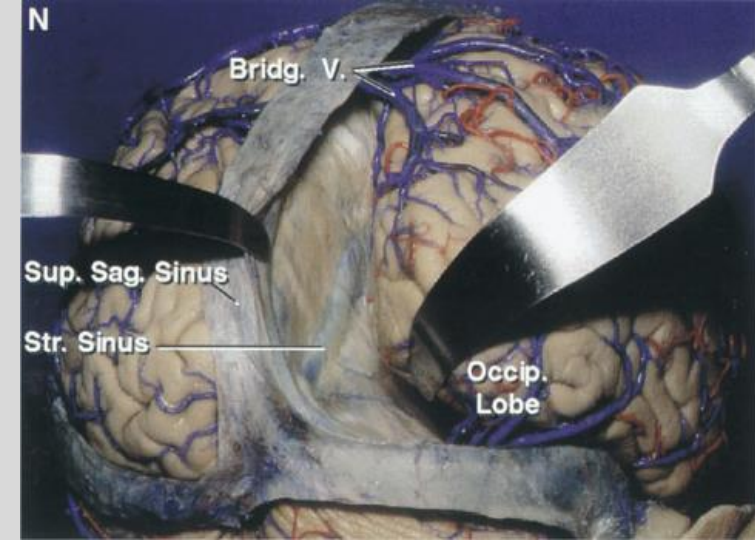
The Dural Folds

Falx cerebri

- Sickle-shaped fold of formed by the meningeal dura mater
- Located in the longitudinal fissure between the cerebral hemispheres
- Contains the Superior Sagittal Sinus and Inferior Sagittal Sinus
- Separates the two cerebral hemispheres, providing structural support and limiting side-to-side movement

Tentorium Cerebelli

- Dural fold that forms a horizontal partition between the cerebrum and cerebellum
- Supports the occipital lobes and prevents compression of the cerebellum, separating the cerebrum from the cerebellum



Rhoton cranial anatomy and surgical approaches

More on this in the Cerebrovascular anatomy lecture..

Brain Herniations

- A brain herniation occurs when something inside the skull produces pressure that moves brain tissues. This is most often the result of brain swelling or bleeding from a head injury, stroke or tumor.

Types:

A. Transtentorial (Uncal) Herniation:

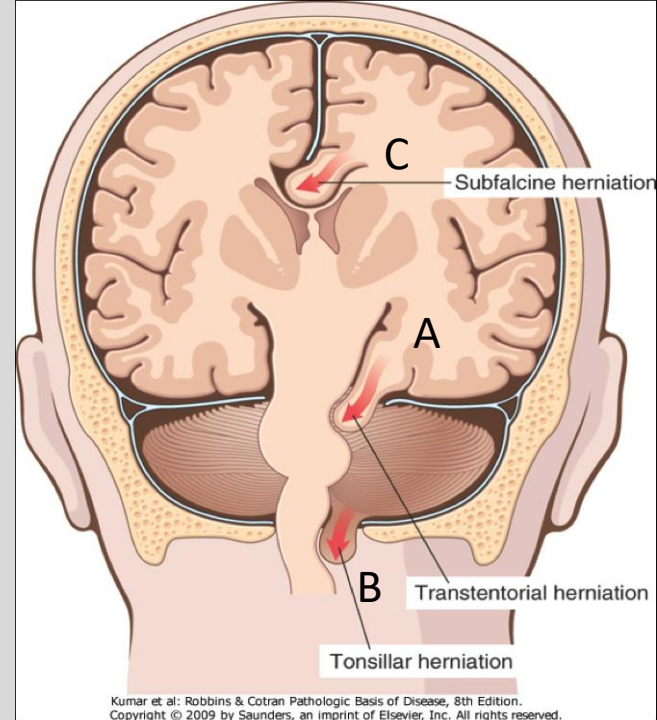
- i. Medial portion of temporal lobe, Uncus, herniates downward through the tentorium cerebelli.

B. Transforaminal (Tonsillar) Herniation

- i. The cerebellar tonsils herniate down through the foramen magnum

C. Subfalcine (Cingulate) Herniation

- i. The cingulate gyrus herniates down below the falx cerebri.



CT brain

More on this in the Cerebrovascular pathology lecture..

Subfalcine Herniation

- Protrusion of cingulate gyrus below the falx
- **Most common** type of herniation.
- Often called “**midline shift**”- the mass effect from the bleed/lesion pushing the brain over the midline.
- Can cause headaches due to **increase intracranial pressure** and **contralateral leg weakness**
- The herniated cingulate gyrus can compress important arteries and further cause ischemia.

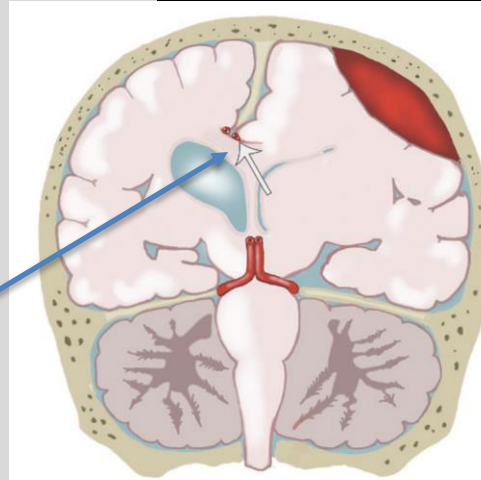
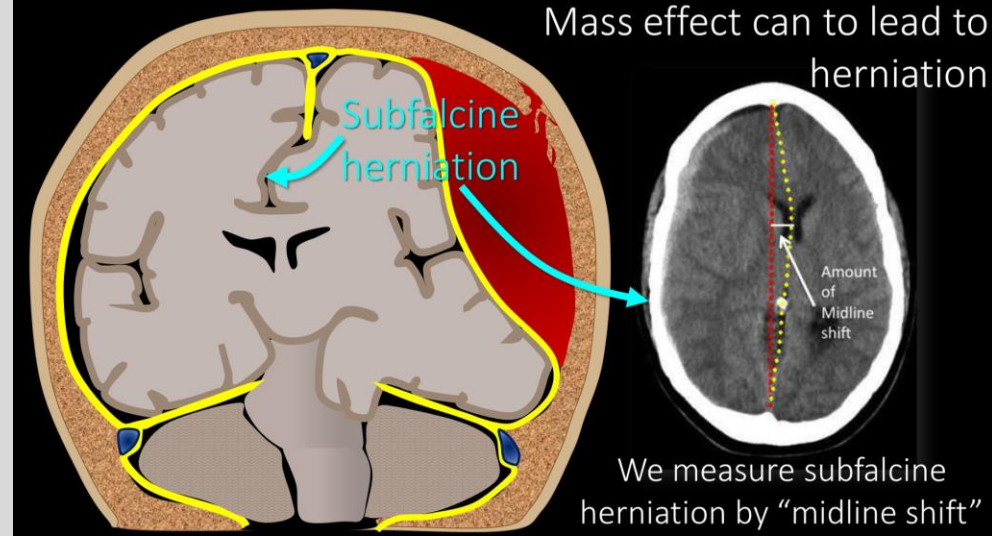
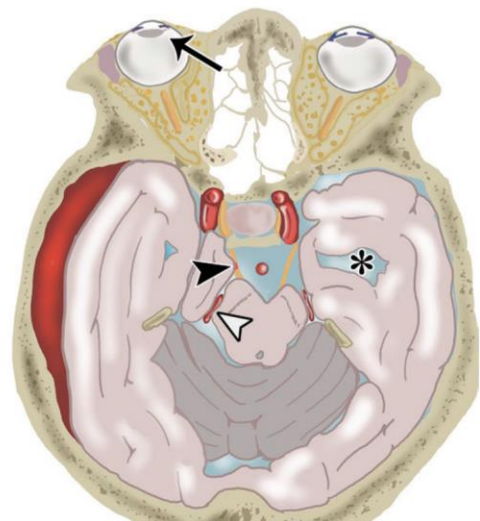
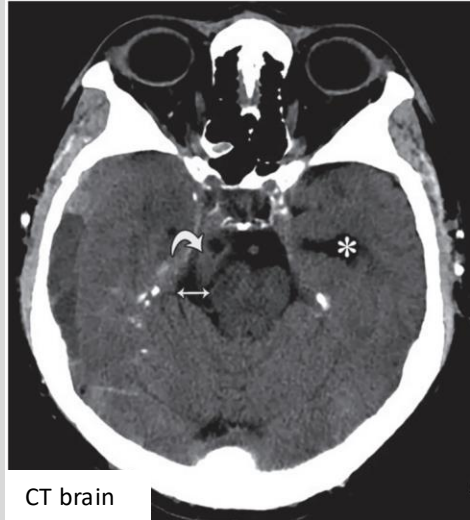


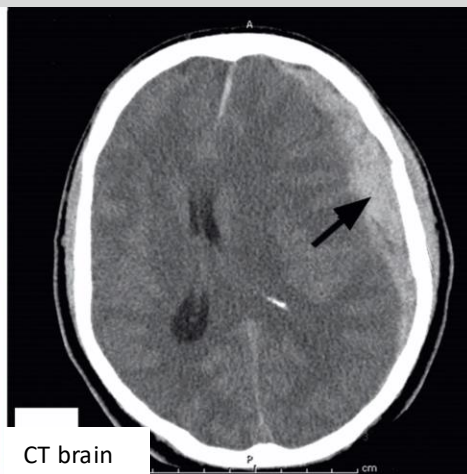
Figure 8. Drawing shows compression of the pericallosal artery (arrow) against the falx cerebri due to a subfalcine hernia.

Transtentorial Herniation

- Transtentorial (Uncal) Herniation: the uncus compresses the **ipsilateral oculomotor nerve**: **ipsilateral fixed and dilated pupil**--> **"blown pupil"**
 - Need immediate neurosurgical intervention**: Craniectomy to evacuation of hematoma and relieve pressure.



Riveros Gilardi B, Muñoz López JL, Hernández Villegas AC, Garay Mora JA, Rico Rodríguez OC, Chávez Appendini R, De la Mora Malváez M, Higuera Galleja JA. Types of Cerebral Herniation and Their Imaging Features. Radiographics. 2019 Oct;39(6):1598-1610. doi: 10.1148/rg.2019190018. PMID: 31589570.

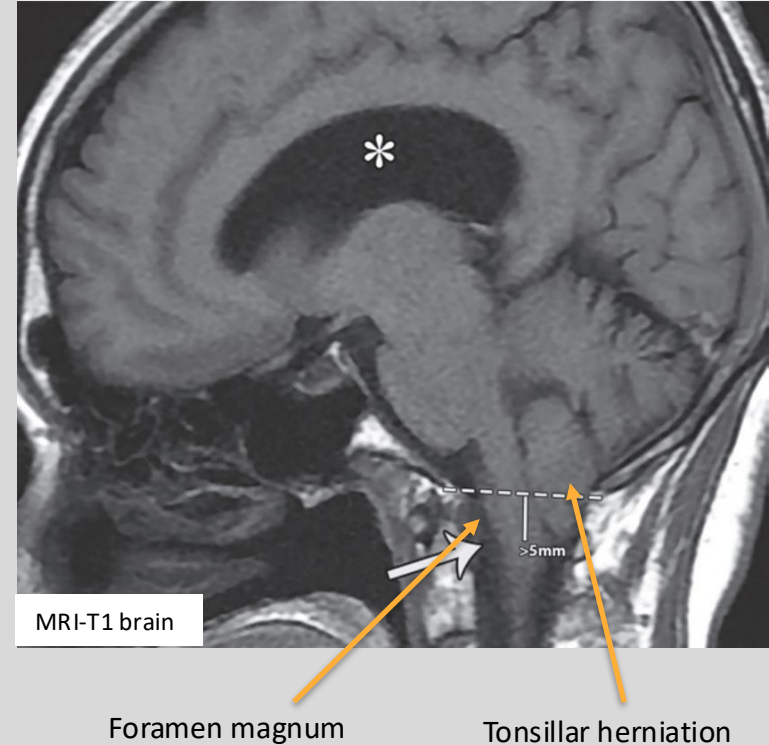


Blown pupil: >5mm

More on this in the Cerebrovascular pathology lecture..

Transforaminal Herniation

- Cerebellar tonsils protrude through the foramen magnum due to increased intracranial pressure, often resulting from conditions like traumatic brain injury, hemorrhage, or tumor.
- **Compression of the brainstem and medulla oblongata, disrupting vital functions such as respiration and cardiovascular regulation.**
- **Obtundation** → Neurosurgical emergency.
- Often very poor prognosis.

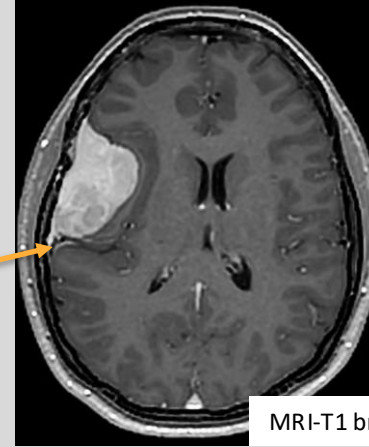


Meningeal lesions and infections

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Meningioma:

- **most common** primary brain tumor
- **benign, well-circumscribed, slow-growing** tumors that arise from the meninges (arachnoid layer)
- 90% of meningiomas are supratentorial
- Classic **dural tail in MRI**



MRI-T1 brain

More on this in the
Cerebrovascular
pathology lecture..

Meningitis:

- **Inflammation of the pia-arachnoid layer** (leptomeninges) of the brain, spinal cord, or both
- Bacterial/Viral/Fungal
- Characterized clinically by **fever, headache, nuchal rigidity, brudzinski's and kernig's signs**
- **Lumbar puncture** → CSF analysis to identify causative agent

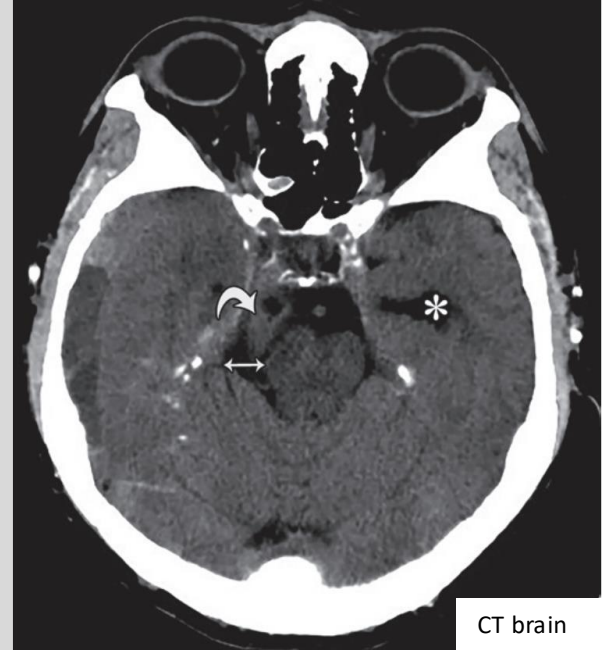


CT +

**Diffuse
leptomeningeal
enhancement on
CT w/ contrast**

Clinical Question

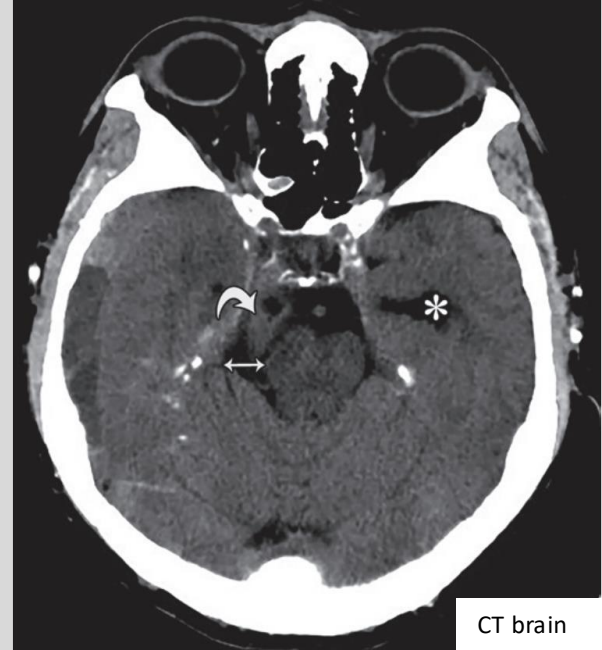
A 26-year-old man with end-stage kidney disease presents with a dilated, unresponsive right pupil (6 mm) shortly after a hemodialysis session. An axial CT scan reveals a subdural hematoma with downward displacement of the medial temporal lobe below a dural fold. What is the most likely diagnosis, and what is the next step in management?



- A.** Cingulate herniation; administer mannitol
- B.** Brainstem infarction; initiate antihypertensive therapy
- C.** Uncal herniation; initiate urgent neurosurgical intervention
- D.** Hydrocephalus; perform ventriculoperitoneal shunt placement

Clinical Question

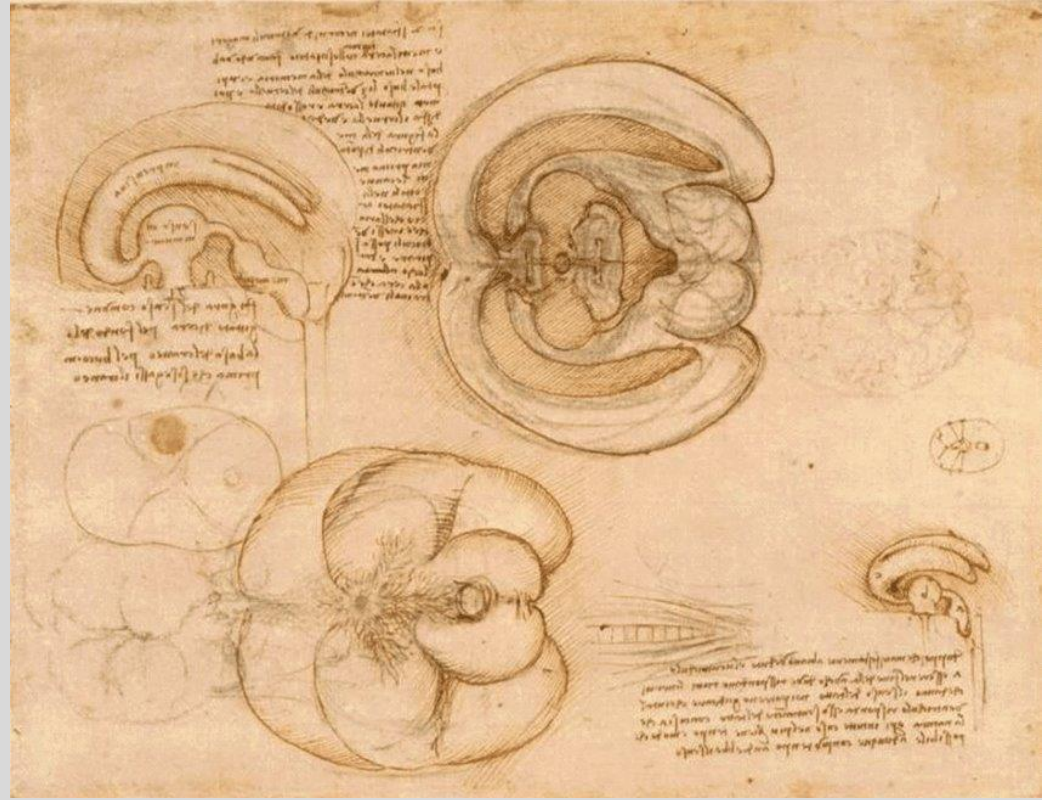
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The patient's dilated, unresponsive pupil and CT findings of downward displacement of the medial temporal lobe below a dural fold, indicate Uncal herniation. This is a neurosurgical emergency requiring immediate intervention to relieve intracranial pressure and prevent life-threatening brainstem compression.

The Ventricular system



Leonardo da Vinci

- 1st person to define and draw the ventricular system
- Ventriculography of brain of an ox: 1508

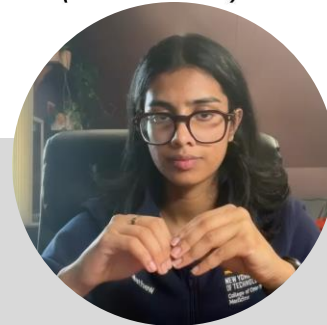
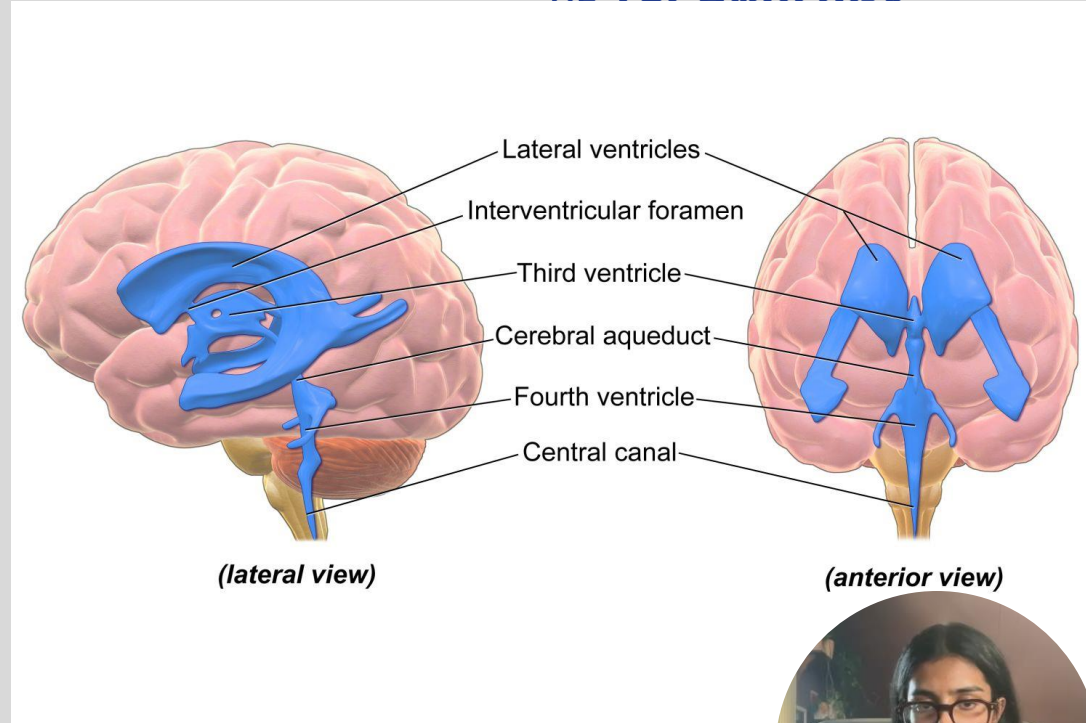
The Ventricular system

But now you will get
your first lesson on
ventricular system from
da Vinci ;)



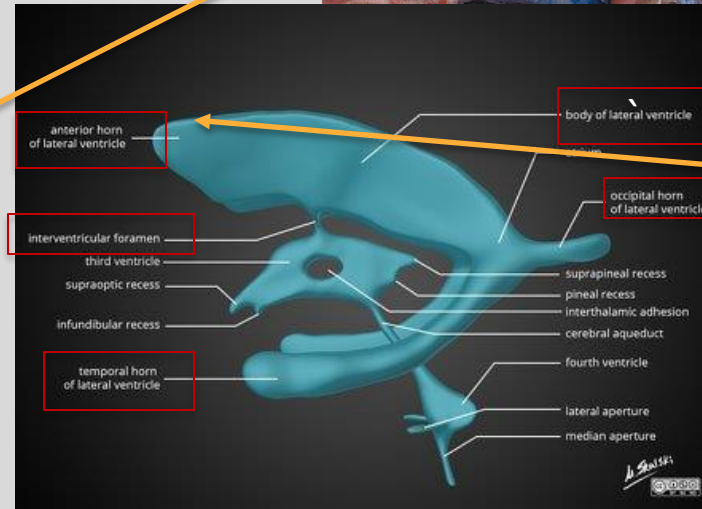
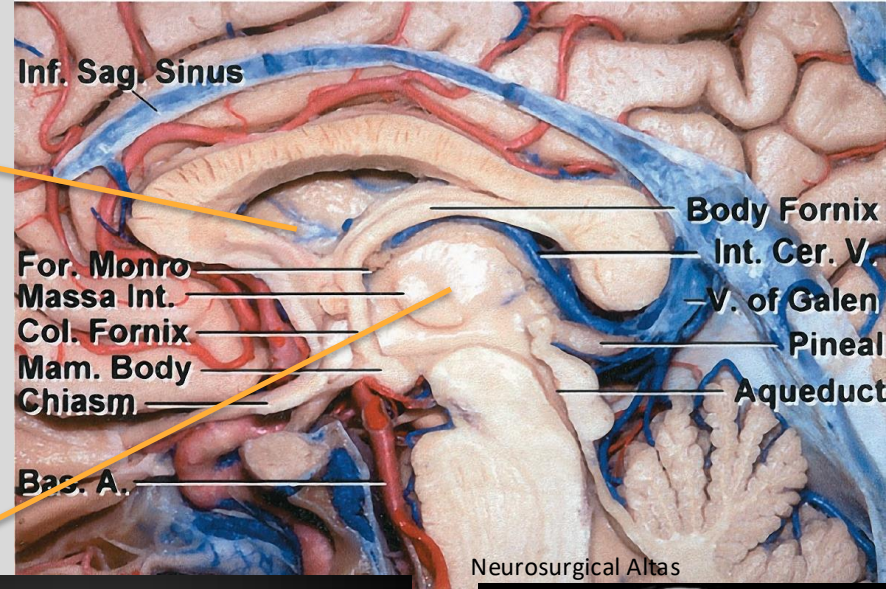
The Ventricular system

- The ventricular system is a network of **interconnected cavities in the brain**, filled with **cerebrospinal fluid (CSF)**, which cushions the brain, removes waste, and **maintains intracranial pressure**.
- Lined by **ependymal cells** which form the blood-CSF barrier.
- **2 Lateral ventricles** : 1 in each cerebral hemisphere
- **Third ventricle** in the diencephalon
- **4th ventricle** in the hindbrain



Lateral Ventricles

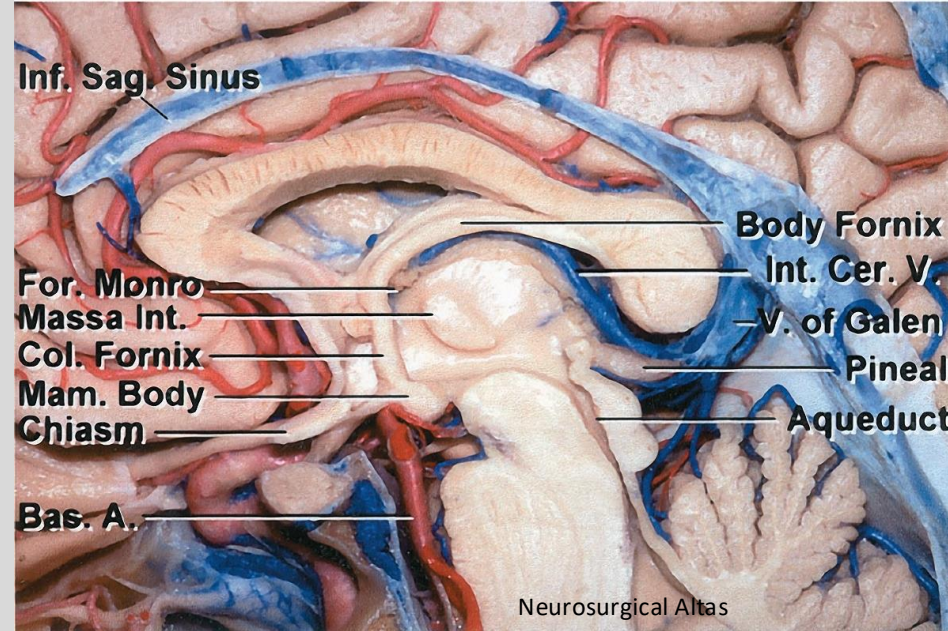
- Located within each cerebral hemisphere.
- **Largest ventricles, with three horns** (frontal, occipital, and temporal).
- Separated by **septum pellucidum**
- Communicates to the **third ventricle** through **2 interventricular foramen (Foramina of Monro)**



CT brain

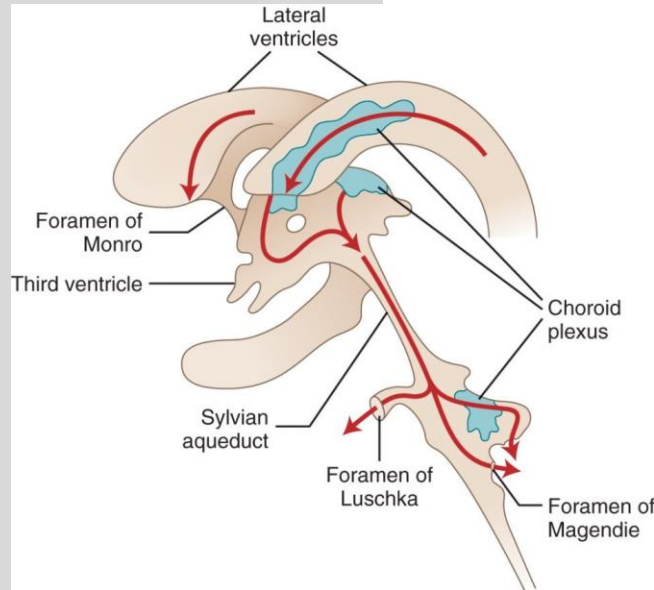
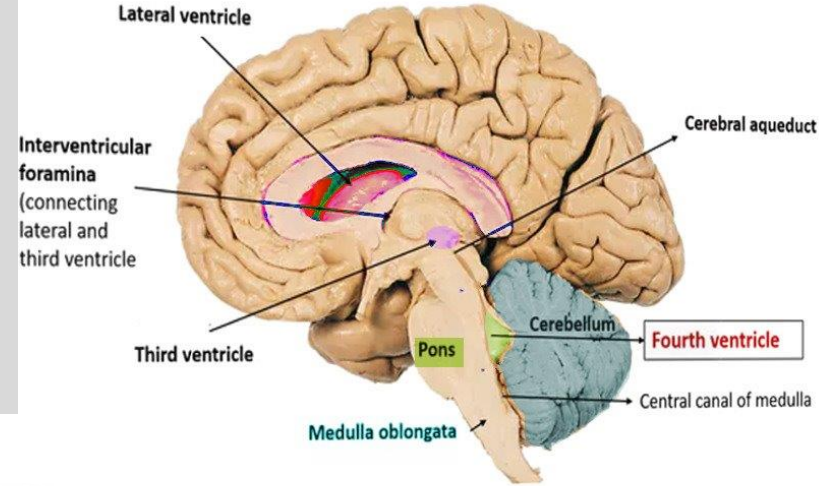
Third Ventricle

- Median slit-like cavity **between the 2 thalami and part of hypothalamus.**
- Communicates with lateral ventricles via the **Foramina of Monro.**
- Communicates with fourth ventricle via the **aqueduct of Sylvius.**
 - Narrowest portion of ventricles
 - Can get obstructed leading to **obstructive hydrocephalus**



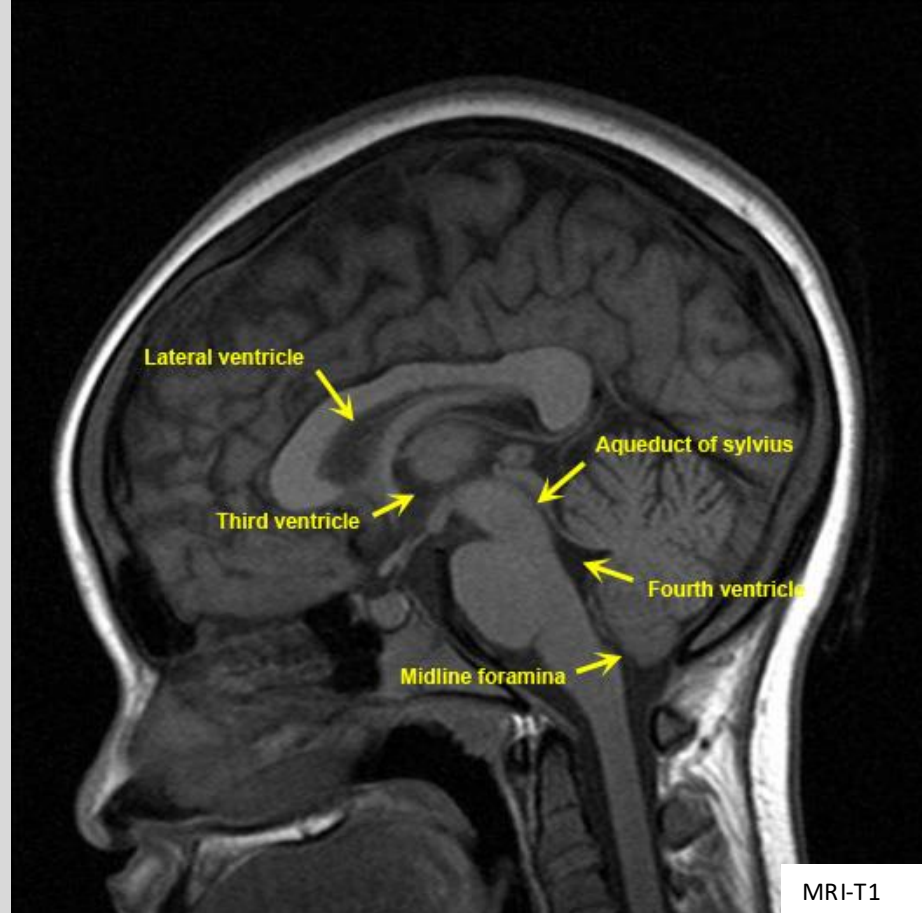
Fourth Ventricle

- Situated between the **brainstem and cerebellum**.
- Connects to the **third ventricle** via the **Cerebral Aqueduct** (Aqueduct of Sylvius).
- **Opens into the subarachnoid space** via the 2 **Foramina of Luschka** (lateral) and **Foramen of Magendie** (medial).



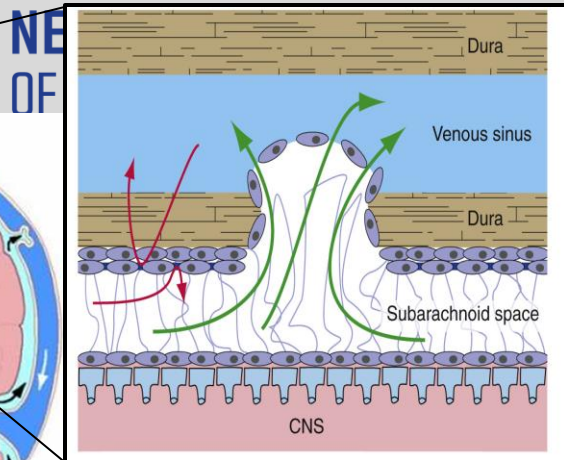
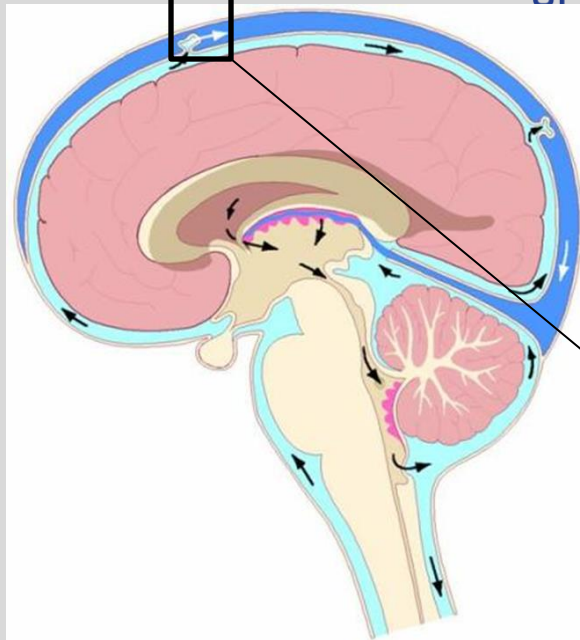
Circulation of CSF

- **Cerebrospinal fluid (CSF)** surrounds the brain and spinal cord, and may function as a shock absorber for the CNS.
- It may also serve an **immunological function** analogous to the lymphatic system (so-called “glymphatic”)
- It is an **ultrafiltrate** of the **plasma** secreted by specialized cells that line the ventricles called the **choroid plexus** (upto 500 ml/day)
- The CSF circulate via **lateral ventricles** → **foramen of Monro** → **third ventricle** → **sylvian aqueduct** → **fourth ventricle** → **foramen of Magendie and foramen of Magendie** → **subarachnoid space** → **arachnoid granulation** → **dural sinuses**.



MRI-T1

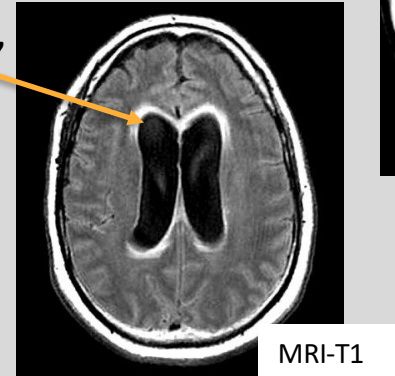
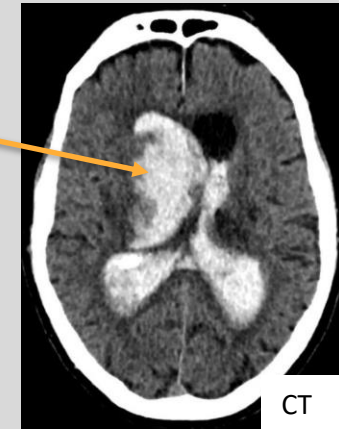
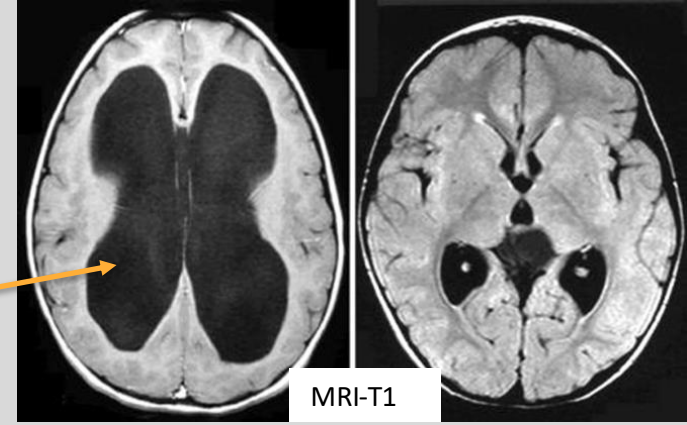
Arachnoid Granulations



Arachnoid villus

Ventricular System Pathologies

- **Choroid plexus tumors:** rare benign tumors (papilloma), can both obstruct flow of CSF and overproduce CSF → **hydrocephalus** and ventriculomegaly
- **Intraventricular hemorrhage:** can result from **extension of intracerebral hemorrhage** or trauma, commonly seen in **preterm infants**.
- **Ventriculitis:** Inflammation of the **ependymal** lining of the cerebral ventricles, usually secondary to an infection such as posttraumatic or implant-related meningitis.



Hydrocephalus

An abnormal accumulation of cerebrospinal fluid within the ventricles of the brain.

- either due to **subnormal CSF reabsorption** or, rarely, **CSF overproduction** (choroid plexus papilloma) .
- Other etiologies: congenital, normal pressure hydrocephalus, idiopathic, ex vacuo etc..

Subnormal CSF reabsorption

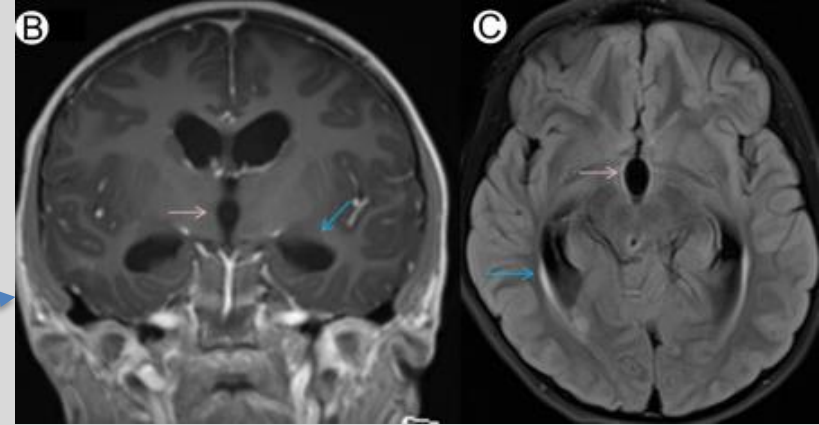
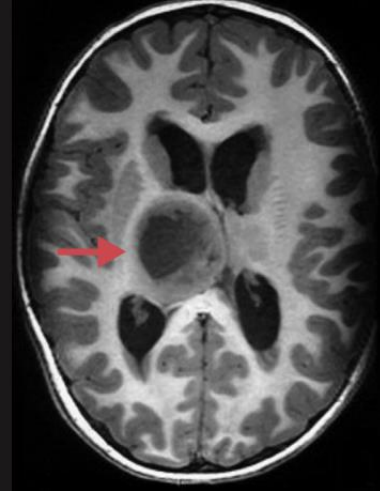
1. **Obstructive hydrocephalus** (AKA non-communicating): block proximal to the arachnoid granulations

- I. On CT or MRI: enlargement of ventricles proximal to block
- II. Causes: tumors, cysts, intraparenchymal or intraventricular hemorrhage

2. **Communicating hydrocephalus** (AKA non-obstructive): defect in CSF reabsorption by the AG

- I. Causes: Meningitis, trauma, subarachnoid hemorrhage (SAH).

MRI-T1: stenosis of aqueduct of Sylvius, colloid cyst blocking foramen of Monro, tumor

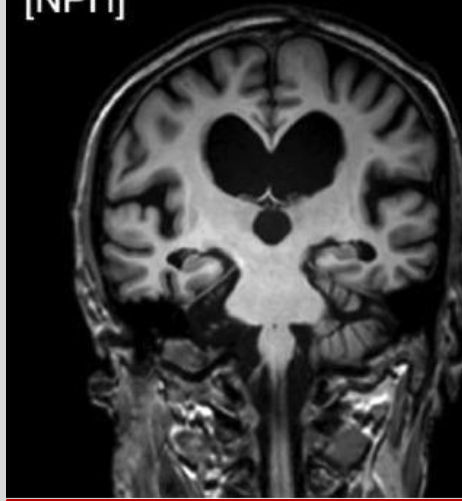


MRI-T1: Post SAH hydrocephalus. Dilatation of third ventricle (yellow arrow) and temporal horn (blue arrow).

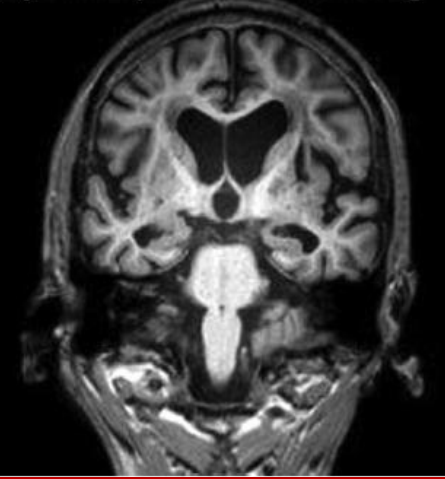
Hydrocephalus

- **Hydrocephalus ex vacuo:** An expansion of the ventricles due to tissue loss. Seen with **Alzheimer's disease**.
- **Normal pressure hydrocephalus (NPH)** seen in older people: hydrocephalus with normal intracranial pressure.
 - Presents with triad of urinary incontinence, cognitive dysfunction and gait abnormality, : **“WET, WACKY, WOBBLY”**
- **Pseudo-tumor cerebri:** (Idiopathic Intracranial Hypertension): seen in **in young females**: characterized by increased intracranial pressure (ICP) without an identifiable mass or obstruction in the brain.
 - Headache, transient visual obscurations, pulsatile tinnitus, and dizziness.

[NPH]



[Hydrocephalus ex vacuo]



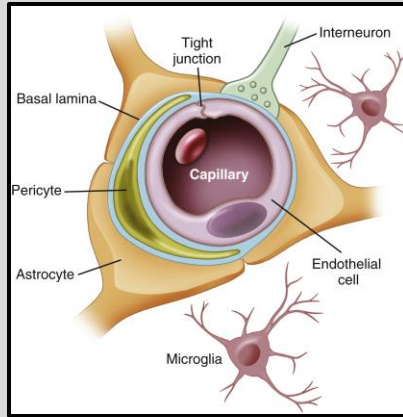
MRI-T1: Left: NPH brain. Right: hydrocephalus ex vacuo in Alzheimer's brain-extensive cortical atrophy- increase in sulci and decreased gyri surface area

Agarwal A, Bathia G, Kanekar S. Imaging of Communicating Hydrocephalus. Semin Ultrasound CT MR. 2016 Apr;37(2):100-8. doi: 10.1053/j.sult.2016.02.007. Epub 2016 Feb 23. PMID: 27063660.

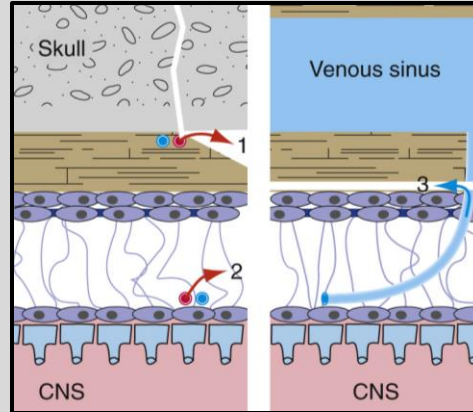
More on this in later pathology lectures..

Summary

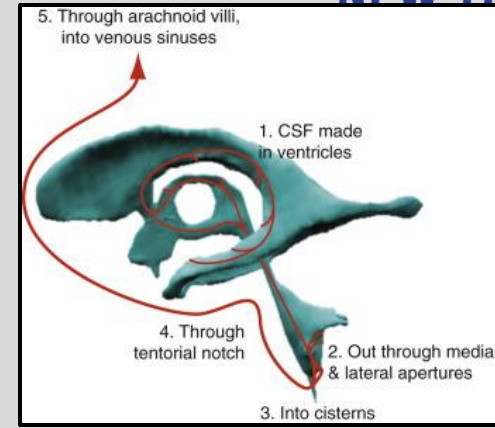
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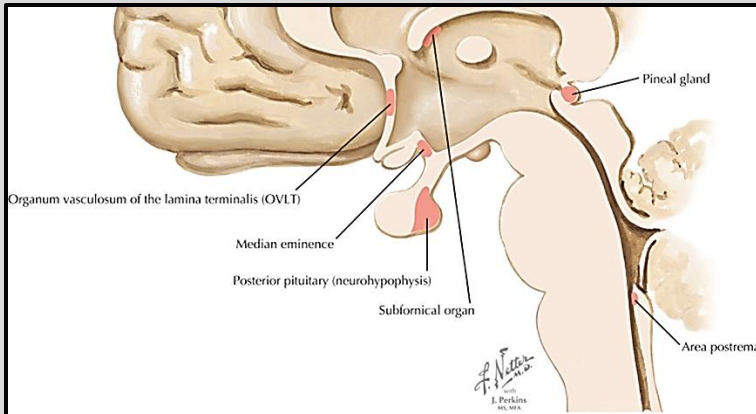
Meninges and meningeal spaces



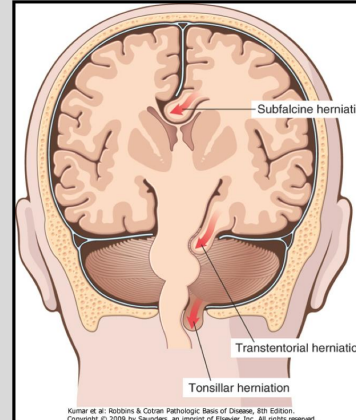
Ventricular system and CSF flow



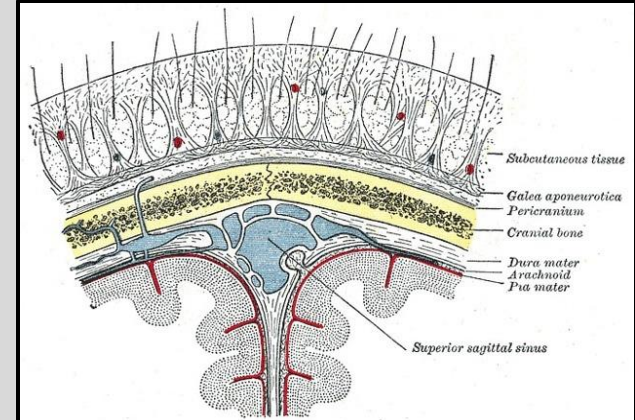
Hydrocephalus



Circumventricular organs



Brain herniations



The scalp and the skull

THANK YOU!

Please email me with
any questions:
vmathe03@nyit.edu



Lecture and Lab Feedback Form:

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